

Assignment D1-V7: Planetary Rover – Solar Energy and Charging Dynamics

Scenario

The rover is equipped with solar panels and can recharge its battery when located in sunlight-exposed regions. Charging efficiency varies depending on location.

The rover must balance exploration and energy recovery.

Domain Characteristics

- Robot: single rover
- Environment: graph with energy zones
- Tasks: exploration, recharging
- Constraints: energy availability

Modelling Guidelines

- Distinguish between energy consumption and energy generation.
- Avoid modelling recharge as instantaneous unless justified.
- Ensure that energy constraints influence planning.

Q1 – Basic PDDL Model

You must:

- Model recharge as an action.
- Provide:
 - one problem where recharge is optional
 - one where recharge is necessary
- Provide valid plans.

Q2 – PDDL+ Model

You must:

- Introduce a continuous charging process.
- Introduce a night-time event reducing or stopping charging.
- Show how timing affects feasibility.

Discussion

Discuss:

- discrete vs continuous energy modelling
- planning under energy uncertainty